FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

PART TEST – III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 264**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Three sections: **Section-A Section-B & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 10, 21 – 30, 41 – 50)** contains 30 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–B (11 – 12, 31 – 32, 51 – 52)** contains 6 Match the following Type questions. Each question having 4 statements in Column I & 5 statements in Column II with any given statement in Column I having correct matching with 1 or more statement (s) given in Column II. Each statement carries **+2 marks** for correct answer and **–1 mark** for wrong answer.

3. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

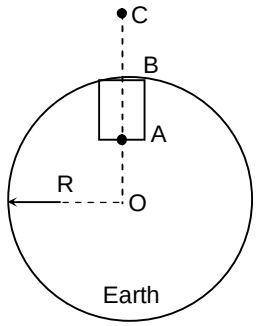
PART – I

SECTION – A

One OR More Than One Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

1. First an object is slowly lifted from the bottom (point – A) of a shaft of depth $h_1 = \frac{R}{2}$ to earth's surface (point-B) and then it is slowly lifted still higher to attain an altitude $h_2 = \frac{R}{2}$ above the earth's surface (Point C). W_1 and W_2 are the work performed in two cases respectively. Choose the correct option(s)



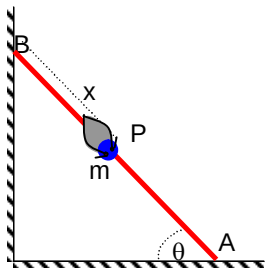
(A) $W_1 > W_2$

(B) $W_1 < W_2$

(C) $\left|\frac{W_1 + W_2}{W_1 - W_2}\right| = 17$

(D) $\left|\frac{W_1 + W_2}{W_1 - W_2}\right| = 9$

2. A light stick of length ℓ rests with its one end against the smooth wall and other end against the smooth horizontal floor as shown in the figure. The bug starts at rest from point B and moves such that the stick always remains at rest. a_P is the magnitude of acceleration of bug of mass m, which depends upon its distance of x from the top end of the stick. Choose the correct option(s)



(A) $a_P = \frac{g}{\sin\theta}\left(1 - \frac{x}{\ell}\right)$

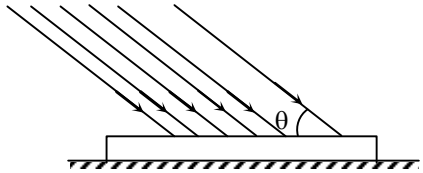
(B) $a_P = \frac{g}{\cos\theta}\left(1 - \frac{x}{\ell}\right)$

(C) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\frac{\pi}{2}\sqrt{\frac{\ell\sin\theta}{3g}}$

(D) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\frac{\pi}{2}\sqrt{\frac{\ell\sin\theta}{g}}$

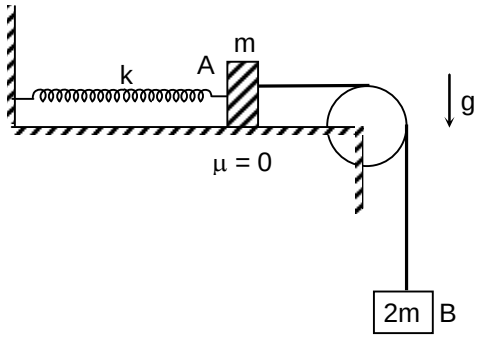
Space for Rough work

3. Monochromatic rays of intensity I are falling on a metal plate of surface area A placed on a rough horizontal surface at certain angle θ as shown in figure. Choose correct statement (s) based on above information:



(A) There is a value of θ for which plate will not move however high the intensity of radiation is
(B) Plate will not move if plate is perfectly reflecting irrespective of the value of intensity.
(C) If rays are falling perpendiculars to surface plate will not move
(D) None of these

4. All the surfaces are frictionless and system is released from rest when ideal spring of stiffness k is in its relaxed state. Choose the correct statement.

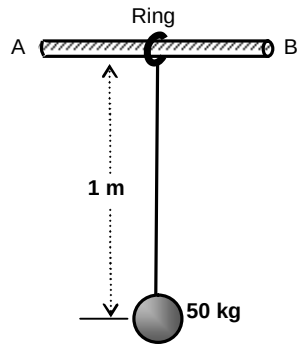


(A) The magnitude of maximum acceleration of particle B is $\frac{g}{3}$
(B) The magnitude of maximum acceleration of particle B is $\frac{2g}{3}$
(C) The maximum speed of particle A is $2g\sqrt{\frac{m}{3k}}$
(D) The maximum speed of particle A is $g\sqrt{\frac{m}{3k}}$

5. We are given the following atomic masses:
$^{238}_{92}U = 238.05079$ u $\quad ^{4}_{2}He = 4.00260$ u
$^{234}_{90}Th = 234.04363$ u $\quad ^{1}_{1}H = 1.00783$ u
$^{237}_{91}Pa = 237.05121$ u
Here the symbol Pa is for the element protactinium (Z = 91).
(A) The energy released during the α-decay of $^{238}_{92}U$ is 4.75 MeV approximately.
(B) The energy released during the α-decay of $^{238}_{92}U$ is 4.25 MeV approximately.
(C) The emission of proton from $^{238}_{92}U$ can be spontaneous.
(D) The emission of proton from $^{238}_{92}U$ can not be spontaneous.

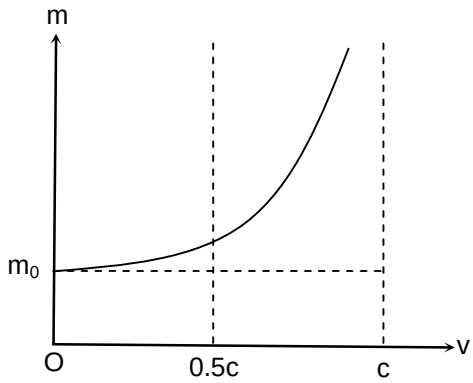
Space for Rough work

6. A 50–kg ball is attached to one end of a 1 m cord that has a mass of 0.10 kg. The other end of the cord is attached to a ring that can slide on a frictionless horizontal shaft AB, as shown in the diagram. A horizontal blow is delivered to the cord and excites the fundamental vibration with a maximum transverse velocity of 15 m/s. Assume that the ball remains essentially stationary as the cord vibrates. [g = 9.8 m/s^2]



(A) The frequency of the fundamental vibration is 17.5 Hz.
(B) The amplitude of the motion is 13.6×10^{-2} m
(C) If the blow is delivered as an impulse to a stationary cord, the wave function for the cord is $y(x, t) = A \sin(\pi x/2L) \sin \omega t$, where the ball is located at x = 0 and positive x-axis is along the string.
(D) The period of the pendulum motion of the hanging ball is 2 sec. approximately.

7. Mass $m = \dfrac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}}$ as a function of velocity is shown in the graph. m_0 is the rest mass of the system. The kinetic energy of system is given as $K = mc^2 - m_c c^2$
Choose correct option(s)



(A) If a system is moving with speed v = 0.5c, the mass of system is increased by 15.5%
(B) Electrons in Cornell university synchrotron reach a velocity of v = 0.8c. The mass of these electrons are approximately $\left(\dfrac{5}{3}\right)$ time the rest mass of electron.
(C) The Bevatron a proton accelerator generates the accelerated protons with kinetic energy 10^{-9} Joules. The mass of accelerated protons is 8.68 times the actual mass of proton approximately $\left(m_{p_0} = 1.67 \times 10^{-27} \text{kg}\right)$
(D) If the kinetic energy of particle is equal to its rest mass energy then speed of particle is 0.75c approximately.

Space for Rough work

8. A rod of length l = 100 cm is fixed at 30 cm from both end. If velocity of transverse wave in rod is v(cms^{-1}), then choose the correct option(s)
(A) Fundamental frequency (in Hz) of transverse wave in rod is v/40
(B) Second overtone in rod will be 5th harmonic
(C) Frequency of third overtone (in Hz) is 7v/40
(D) Third overtone is 5th harmonic.

9. If voltage applied to an X-ray tube increased from V = 10 kV to 20 kV. The wave length interval between K_α-line and short wave cut off of continuous X-ray increases by factor of 3 $\frac{Rhc}{e} = 13.6\,V$, where R is Rydberg constant, h is plank's constant, c is speed of light in vacuum and e is charge on electron.
(A) Atomic no. of target metal used is 29
(B) Cut-off wavelength when V = 10 kV is 1.2 Å
(C) Cut off wavelength when V = 10 kV is 2 Å
(D) Atomic no. of target metal used is 26

10. Consider a 20 W bulb emitting light of wavelength 5000 Å and shining on a metal surface kept at a distance 2m. Assume that the metal surface has work function of 2 eV and that each atom on the metal surface can be treated as a circular disk of radius 1.5 Å.
(A) Number of photon emitted by bulb is 5×10^{19} per second
(B) Time required by atomic disk to receive energy equal to work function of metal is 22.4 sec.
(C) Number of photon received by disk to receive 2 eV energy is 4
(D) Number of photons emitted by bulb is 5×10^{17} per sec.

Space for Rough work

SECTION – B

(Matching Type)

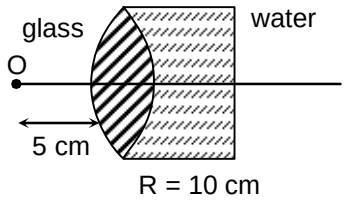
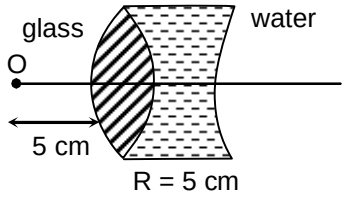
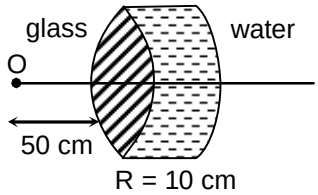
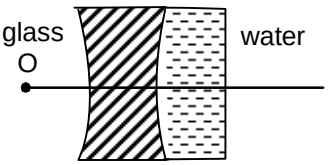
This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct

11. In the photoelectric effect experiment, if f is the frequency of radiations incident on the metal surface and I is the intensity of the incident radiations, then match the following columns.

Column -I		Column -II	
(A)	If f is increased keeping I and work-function constant	(p)	Stopping potential increases
(B)	If distance between cathode and anode is increased	(q)	Saturation current increases
(C)	If I is increased keeping f and work-function constant	(r)	Maximum kinetic energy of photoelectron increases
(D)	Work-function is decreased keeping f and I constant	(s)	Stopping potential remains same
		(t)	Stopping potential decreases

Space for Rough work

12. Two thin lens made of glass $\left(\mu = \frac{3}{2}\right)$ and water $\left(\mu = \frac{4}{3}\right)$ kept in contact with each other. An object is placed at point O. Assume all curved surfaces have equal radius.
Match the following Columns:

Column – I		Column - II	
(A)		(p)	Image of O will be Real
(B)		(q)	Image of O will be Virtual
(C)		(r)	Focal length = 10 cm
(D)		(s)	Focal length = 15 cm
		(t)	Focal length = 25 cm

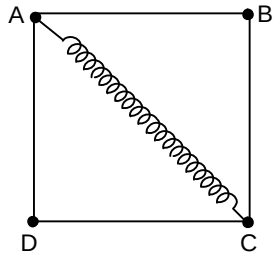
Space for Rough work

SECTION – C

(One Integer Value Correct Type)

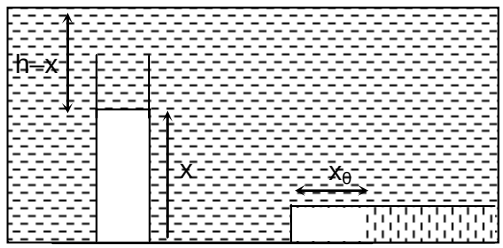
This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

13. A square hinged-structure (ABCD) is formed with four massless rods which lies on the smooth horizontal table. The hinge-D of structure is fixed with table and hinge A, hinge-B and hinge-C are free to move on the table. A point mass m is attached at hinge-B, and an ideal spring of stiffness k is connected between hinge-A and hinge-C as shown in the figure. System is in equilibrium. The mass m is slightly displaced along line BD and released to perform SHM. The time period of SHM is $\pi\sqrt{\frac{nm}{k}}$. Find



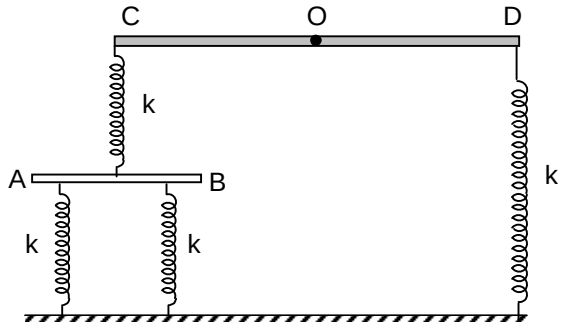
14. The mean pressure is $\frac{6K}{5}$, which rain renders to vertical windshield of automobile, moving with constant velocity of magnitude v = 12 m/s. Consider that raindrops fall vertically with speed u = 5 m/s. The intensity of rainfall deposits h = 2 cm of sediments in time τ = 1 minute. [$\rho = 10^3$ kg/m^3 is the density of liquid] (Assume collisions are inelastic). Calculate K.

15. On the bottom of lake at a depth h = 100m, a horizontal pipe of length 80m with piston is kept as shown in the figure. The piston is light and movable. Between the piston and pipe some air is captured (x_0 = 9m). The pipe is slowly raised to vertical position and open end upward. The length of air column (x) in SI units is 5K. Find value of K. (Disregard the atmospheric pressure & piston and pipe are air tight. Temperature of gas remains constant).

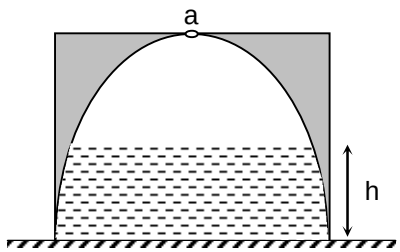


Space for Rough work

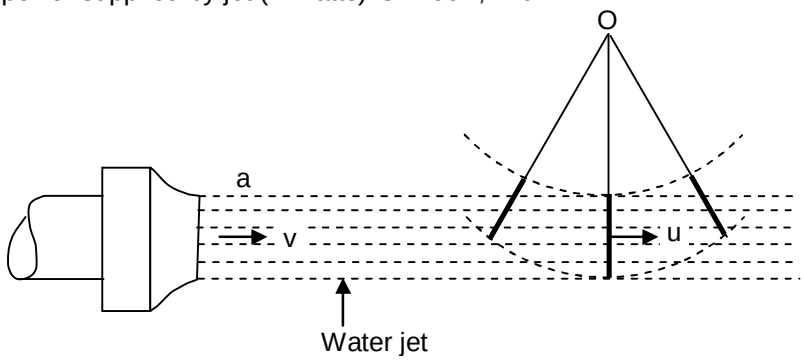
16. In the figure shown rod AB is light and rigid while rod CD is also rigid and hinged at midpoint O Rod can rotate without friction about O. The time period of small oscillation of rod CD is $T = 2\pi\sqrt{\frac{m}{nk}}$. Where m is mass of rod CD and k is spring constant of each spring. The value of n is



17. Consider a cuboidal vessel (2R × 2R × R) with a hemispherical cavity of radius R, kept at a horizontal smooth surface as shown in the figure. The vessel has very small hole of cross-sectional area **"a"**. Now water of density ρ is poured into space developed due horizontal surface and vessel through hole very slowly. When the height of water level is h, the vessel lifts off the surface and liquid leaks through space generated. (R = 2m, $\rho = 10^3$ kg/m^3, m = 9π kg). If the value of h is 5k cm, find the valve of k.
(Atmospheric pressure = 10^5 N/m^2).



18. In an undershot water wheel, the cross-sectional area $a = 0.1 m^2$ of the stream is striking the series of radial flat vanes of the wheel. The velocity of stream is 6 m/s. The velocity of vanes is 3 m/s. If the power supplied by jet (in watts) is 2700K, find K.



19. A man of height = 3/2 m, wants to see himself in plane mirror from top to bottom. The plane mirror is inclined with vertical wall at angle α = 53º. If the least size of mirror to see him is $\frac{3}{n}$m. the distance of eye from mirror is d = 3m, find the value of n.

20. If ε_0, B, V represent permittivity of free space, magnitude of magnetic field and volume of space respectively, then the dimension of $\varepsilon_0 B^2 V$ is $\left[M^a L^b T^c\right]$. Find a + b + c.

Space for Rough work

Chemistry

PART – II

SECTION – A

One OR More Than One Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

21. Select the correct statement
(A) Viscosity in gases arises principally from the molecular diffusion that transport momentum between layers of flow. The net effect will be decrease in the relative rate of movement of one layer with respect to the other.
(B) Viscosity of ideal gas increases as temperature increases.
(C) Viscosity of liquids decreases with the rise in temperature.
(D) At critical temperature the surface of separation between liquid and gas disappear and their viscosity is same.

22. An ideal gas whose adiabatic exponent γ is expanded according to the law $P = \alpha V$, where α is a constant. The initial volume of the gas is equal to V_o. As a result of expansion, the volume increases 4 times. Select the correct option for above information.
(A) Molar heat capacity of gas in the process is $\frac{R(\gamma+1)}{2(\gamma-1)}$
(B) Work done by the gas is $15V_o^2\frac{\alpha}{2}$
(C) Change in internal energy of gas is $15V_o^2\frac{\alpha}{2}$
(D) Both B and C are correct

23. Which of the following complexes exists in facial and meridional forms?
(A) $[Co(dien)_2]^{3+}$ (B) $[Co(NH_3)_3Cl_3]$
(C) $[Co(en)_3]^{3+}$ (D) $[Co(gly)_3]$

24. Select the correct statement, for non stoichiometric cuprous oxide $Cu_{1.8}O$.
(A) % of Cu^{2+} in total copper is 11.11% (B) % of Cu^{1+} in total copper is 11.11%
(C) It behaves like p-type semiconductor (D) Defect is metal deficiency defect

25. Which of the following are colourless?
(A) Ce^{3+} (B) La^{3+}
(C) Lu^{3+} (D) Gd^{3+}

Space for Rough work

26. One mole of component P and two moles of component Q are mixed at 27°C to form an ideal binary solution. Select the correct option for given information?

(A) $\Delta S_{mix} = 3.82 \text{ cal } k^{-1}$ (B) $\Delta H_{mix} = 0$

(C) $\Delta S_{mix} = 0$ (D) $\Delta G_{mix} = -1145.7$ cal

27. Select the correct statement for lead storage battery:

(A) It is a reversible cell.
(B) Salt bridge is not required for lead storage battery.
(C) It is recharged by using DC source of current.
(D) Specific gravity of sulphuric acid solution decreases during discharging.

28. Which of the following statements are true for froth floatation process (for the concentration of sulphide ores)?

(A) Pine oil used as collectors.
(B) Cresol enhance non wettability of the mineral particles.
(C) NaCN selectively prevents PbS from coming to the froth but allows ZnS to come with the froth.
(D) The mineral particles become wet by oils while the gangue particles by water.

29. Which of the following elements can give NO_2 as a by product on reaction with conc. HNO_3?

(A) C (B) Cu

(C) Zn (D) Sn

30. Chlorine has great affinity for hydrogen. It reacts with compounds containing hydrogen to form HCl. Which of the following reaction/s give/s HCl?

(A) $H_2O + Cl_2 \longrightarrow$ (B) $H_2S + Cl_2 \longrightarrow$

(C) $C_{10}H_{16} + 8Cl_2 \longrightarrow$ (D) $\underset{(\text{excess})}{NH_3} + 3Cl_2 \longrightarrow$

Space for Rough work

SECTION – B

(Matching Type)

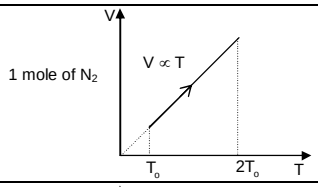
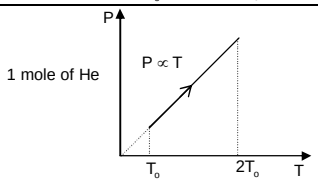
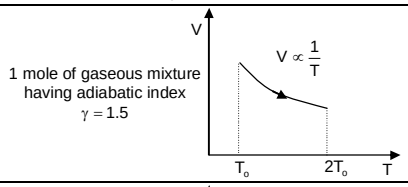
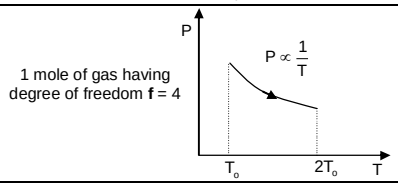
This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct

31. Match the following column of precipitate/mass listed in Column I with the reagent (s) listed in Column II:

Column – I (Observations)		Column – II (Reagents)	
(A)	Mg^{2+} gives pale pink mass with	(p)	NaOH solution
(B)	Pb^{2+} gives yellow ppt. with	(q)	H_2S gas
(C)	Ag^{+} gives black/brown ppt. with	(r)	K_2CrO_4 solution
(D)	Hg_2^{2+} gives black ppt. with	(s)	KI (not in excess) or KI in excess
		(t)	Cobalt nitrate in charcoal cavity test

Space for Rough work

32. In Column I, certain thermodynamic process are given and in Column II, the value of physical quantities are given. Match the Column I with Column II suitably. (Given : ρ = density of gas)

Column – I		**Column – II**	
(A)		(p)	$\lvert Q \rvert \geq 2RT_o$
(B)		(q)	$\lvert \Delta U \rvert \geq \frac{3}{2}RT_o$
(C)		(r)	$\lvert W \rvert \geq RT_o$
(D)		(s)	ρ vs T graph for the process is a straight line
		(t)	P vs ρ graph for the process is a parabola

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

33. How many types of isomerism is exhibited by the complex $[Co(NH_3)_4(NO_2)_2]Cl$?

34. How many millilitres of 0.05 M $K_4[Fe(CN)_6]$ solution is required for titration of 60 ml of 0.01 M $ZnSO_4$ solution, when the product of reaction is $K_2Zn_3[Fe(CN)_6]_2$?

35. A molecule A_x dissolve in water and is non volatile. A solution of certain molality showed a depression of 0.93 K in freezing point. The same solution boiled at 100.26°C. When 7.87 g of A_x was dissolved in 100 g of water, the solution boiled at 100.44°C. Given K_f for water = 1.86 K kg mol^{-1}, atomic mass of A = 31 u. Assume no association or dissociation of solute. Calculate the value of x…….

36. What volume of air (in m^3) is needed for the combustion of 1 m^3 of a gas having the following composition in percentage volume : 2% of C_2H_2, 8% of CO, 35% of CH_4, 50% of H_2 and 5% of non-combustible gas. The air contains 20.8% (by volume) of oxygen.

Space for Rough work

37. Find the total number of metals, which makes a thin protective layer of its oxide on treatment with conc. HNO_3.
Zn, Cu, Al, Pt, Cr, Au, Pb, Fe

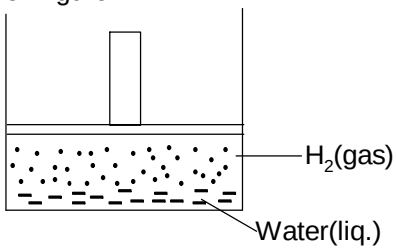
38. The number of salts formed by telluric acid when treated with strong base NaOH is/are….

39. If 6.53×10^{-2} g of metallic zinc is added to 100 ml saturated solution of AgCl, it reacts with Ag^+ of solution as following reaction

$$Zn(s) + 2Ag^+(aq) \rightleftharpoons Zn^{2+}(aq) + 2Ag(s)$$

and approximately 10^{-x} moles of Ag will be precipitated. Calculate the value of x……(Given $E^o_{Zn^{++}/Zn} = -0.76\,V$ $E^o_{Ag^+/Ag} = 0.8\,V$, K_{sp} of AgCl = 10^{-10}, atomic mass of Zn = 65.3u, $10^{52.8813} = 7.61 \times 10^{52}$)

40. Initial volume of H_2 gas saturated with water vapour is confined under a piston in a container is 10 litres as shown in the given figure:



the container also contains some liquid water. The total pressure over liquid water is 80 cm of Hg. If now the piston is removed such that volume of container is doubled, then final total pressure over liquid water in the container is P. The vapour pressure of water is 20 cm of Hg and volume of liquid is negligible. Calculate P/10…….

Space for Rough work

Mathematics

PART – III

SECTION – A

One OR More Than One Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

41. If $D_1, D_2, D_3, ... D_{1000}$ are 1000 doors and $P_1, P_2, P_3, ... P_{1000}$ are 1000 persons. Initially all doors are closed. Changing the status of doors means closing the door if it is open or opening it if it is closed. P_1 changes the status of all doors. Then P_2 changes the status of $D_2, D_4, D_6, ... D_{1000}$ (doors having numbers which are multiples of 2). Then P_3 changes the status of $D_3, D_6, D_9, ... D_{999}$ (doors having number which are multiples of 3) and this process is continued till P_{1000} changes the status of D_{1000}, then the doors which are finally open is/are
(A) D_{961} (B) D_{269}
(C) D_{413} (D) D_{729}

42. Which of the following options are correct?
(A) $[({}^nC_0 + {}^nC_3 + {}^nC_6 + ...) - \frac{1}{2}({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + ...)]^2 + \frac{3}{4}({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + ...)^2 = 1$
(B) If a and b are two positive numbers such that $a^5b^2 = 4$ then the maximum value of $\log_{2^{1/5}}\left(a^2\right)\cdot\log_{2^{1/2}}\left(b^2\right)$ is equal to 4
(C) Constant term in $((((((x-2)^2-2)^2-2)^2-2)^2-2)^2 ..... 2)^2$ is equal to 2
(D) The coefficient of x^{24} in $\left(\frac{{}^{25}C_1}{{}^{25}C_0} - x\right)\left(x - 2^2\frac{{}^{25}C_2}{{}^{25}C_1}\right)\left(x - 3^2\frac{{}^{25}C_3}{{}^{25}C_2}\right)\left(x - 4^2\frac{{}^{25}C_4}{{}^{25}C_3}\right).....$
$\left(x - 25^2\frac{{}^{25}C_{25}}{{}^{25}C_{24}}\right)$ is equal to 2925

43. x_1, x_2, x_3 are three real numbers satisfying the system of equations
$x_1 + 3x_2 + 9x_3 = 27$, $x_1 + 5x_2 + 25x_3 = 125$ and $x_1 + 7x_2 + 49x_3 = 343$, then which of the following options are correct
(A) number of divisors of $x_1 + x_3$ is 16 (B) $\frac{x_1 + x_2}{2}$ is a prime number
(C) $x_3 - x_2$ is a prime number (D) $x_1 + x_2 + x_3$ is square of an integer

Space for rough work

44. $a_1, a_2, a_3, ...$ are distinct terms of an A.P. We call (p, q, r) an increasing triad if a_p, a_q, a_r are in G.P. where p, q, r $\in$ N such that p < q < r. If (5, 9, 16) is an increasing triad, then which of the following option is/are correct
(A) if a_1 is a multiple of 4 then every term of the A.P. is an integer
(B) (85, 149, 261) is an increasing triad
(C) if the common difference of the A.P. is $\frac{1}{4}$, then its first term is $\frac{1}{3}$
(D) ratio of the (4k + 1)th term and $4k^{th}$ term can be 4

45. If $z_1, z_2, z_3\ z_4$ are complex numbers in an Argand plane satisfying $z_1 + z_3 = z_2 + z_4$. A complex number 'z' lies on the line joining z_1 and z_4 such that $\text{Arg}\left(\frac{z-z_2}{z_1-z_2}\right) = \text{Arg}\left(\frac{z_3-z_2}{z-z_2}\right)$. It is given that $|z - z_4| = 5$, $|z - z_2| = |z - z_3| = 6$ then
(A) area of the triangle formed by z, z_1, z_2 is $3\sqrt{7}$ sq. units
(B) area of the triangle formed by z, z_3, z_4 is $\frac{15\sqrt{7}}{4}$ sq. units
(C) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken in order is $\frac{27\sqrt{7}}{2}$ sq. units
(D) area of the quadrilateral formed by the points z_1, z_2, z_3, z_4 taken in order is $\frac{27\sqrt{7}}{4}$ sq. units

46. Which of the following is/are true?
(A) $100^{300} < 300!$
(B) $\sqrt{300^{300}} < 300!$
(C) $100^{300} > 300!$
(D) $\sqrt{300^{300}} > 300!$

47. Which of the following is/are correct?
(A) If A is a n $\times$ n matrix such that $a_{ij} = \left(i^2 + j^2 - 5ij\right)\cdot\left(j - i\right)\ \forall$ i and j then trace (A) = 0
(B) If A is a n $\times$ n matrix such that $a_{ij} = \left(i^2 + j^2 - 5ij\right)\cdot\left(j - i\right)\ \forall$ i and j then trace (A) $\neq$ 0
(C) If P is a 3 $\times$ 3 orthogonal matrix, α, β, γ are the angles made by a straight line with OX, OY, OZ and $A = \begin{bmatrix} \sin^2\alpha & \sin\alpha\cdot\sin\beta & \sin\alpha\cdot\sin\gamma \\ \sin\alpha.\sin\beta & \sin^2\beta & \sin\beta\cdot\sin\gamma \\ \sin\alpha\cdot\sin\gamma & \sin\beta\cdot\sin\gamma & \sin^2\gamma \end{bmatrix}$ and $Q = P^TAP$, then $PQ^6P^T = 32A$
(D) If matrix $A = \left[a_{ij}\right]_{3\times3}$ and matrix $B = \left[b_{ij}\right]_{3\times3}$ where $a_{ij} + a_{ji} = 0$ and $b_{ij} - b_{ji} = 0\ \forall$ i and j then A^6B^7 is a singular matrix

Space for rough work

48. The vertices of a triangle ABC are A ≡ (2, 0, 2), B(–1, 1, 1) and C ≡ (1, –2, 4). The points D and E divide the sides AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that the perpendicular drawn from F to the plane containing ΔABC, meets the plane at the point of intersection of the line segments CD and BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}$ units, then

(A) the volume of the tetrahedron ABCF is $\frac{7}{3}$ cubic units

(B) the volume of the tetrahedron ABCF is $\frac{7}{6}$ cubic units

(C) one of the equation of the line AF is $\vec{r} = \left(2\hat{i} + 2\hat{k}\right) + \lambda\left(2\hat{k} - \hat{i}\right)$ ($\lambda \in R$)

(D) one of the equation of the line AF is $\vec{r} = \left(2\hat{i} + 2\hat{k}\right) + \mu\left(\hat{i} + 7\hat{k}\right)$

49. The direction cosines of two lines are connected by the relations $\ell + m + n = 0$ and $mn = 2\ell(m + n)$, then

(A) $\frac{\ell_1}{\ell_2} + \frac{m_1}{m_2} + \frac{n_1}{n_2}$ is equal to $-\frac{3}{2}$

(B) $\ell_1\ell_2 + m_1m_2 + n_1n_2$ is equal to $-\frac{1}{2}$

(C) $\ell_1m_1n_1 + \ell_2m_2n_2$ is equal to $-\frac{\sqrt{2}}{3\sqrt{3}}$

(D) $(\ell_1 + \ell_2)(m_1 + m_2)(n_1 + n_2)$ is equal to $\frac{1}{3\sqrt{6}}$

50. Let the equation of a straight line L in complex form be $a\bar{z} + \bar{a}z + b = 0$, where a is a complex number and b is a real number, then

(A) the straight line $\frac{z - c}{a} + \frac{i(\bar{z} - \bar{c})}{\bar{a}} = 0$ makes an angle of 45º with L and passes through a point c (where c is a complex number)

(B) the straight line $\frac{z - c}{a} = \frac{i(\bar{z} - \bar{c})}{\bar{a}}$ makes an angle of 45º with L and passes through a point c (where c is a complex number)

(C) the complex slope of the line L is $-\frac{a}{\bar{a}}$

(D) the complex slope of the line L is $\frac{a}{\bar{a}}$

Space for rough work

SECTION – B

(Matching Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct

51. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	If the polynomial $x^3 + ax^2 + bx + c$ is divisible by $(x^2 + 1)$, where a, b, c belong to {1, 2, 3, 4, ..., 10} then a + b + c may be equal to	(p)	6
(B)	A and B are 3×3 matrices of real numbers, where A is symmetric a matrix and B is a skew symmetric a matrix, and $(A + B)(A – B) = (A – B)(A + B)$. If $(AB)^T = (–1)^k AB$ then k may be equal to	(q)	3
(C)	Sum of the digits of $(100^{50} – 43)$ is divisible by	(r)	9
(D)	Let $\vec{a} = (\cos\theta)\hat{i} - (\sin\theta)\hat{j}$, $\vec{b} = (\sin\theta)\hat{i} + (\cos\theta)\hat{j}$, $\vec{c} = \hat{k}$, $\vec{r} = 7\hat{i} + \hat{j} + 10\hat{k}$, if $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$ then the value of $\frac{x^2 + y^2}{z}$ is less than	(s)	10
		(t)	11

52. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	If x and y are two integers such that $289 – x^2 + y^4 = 0$ then the possible value(s) of unit digit of x + 12y + 4 is/are	(p)	1
(B)	If P, Q, R, S be four points in space satisfying $\left\|\overrightarrow{PQ}\right| = 3$, $\left\|\overrightarrow{QR}\right| = 7$, $\left\|\overrightarrow{RS}\right| = 11$, $\left\|\overrightarrow{SP}\right| = 9$, then the value of $\frac{\overrightarrow{PS} \cdot \overrightarrow{QR}}{9}$ is less than	(q)	2
(C)	The sum of three positive integer is 20. If the probability that they form the sides of a triangle is P then 19P is equal to	(r)	3
(D)	The first term of an infinite geometric series is 21. The second term and the sum of the series are both positive integers. If the value of second term is k then the possible value(s) of $\|k – 15\|$ is/are	(s)	4
		(t)	5

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

53. If $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the roots of $x^4 + 2x^3 + bx^2 + cx + d = 0$ such that $\alpha_1 - \alpha_3 = \alpha_4 - \alpha_2$, then $b - c$ is equal to _____

54. A chess match between two players A and B is won by whoever first wins a total of two games. Probability of A's winning, drawing and losing any particular game are $\frac{1}{6}$, $\frac{1}{3}$ and $\frac{1}{2}$ respectively. (The games are independent). If the probability that B wins the match in the 4th game is p, then 6p is equal to _____

55. If k_1 and k_2 ($k_1 > k_2$) are two non-zero integral values of k for which the cubic equation $x^3 + 3x^2 + k = 0$ has all integer roots, then the value of $k_1 - k_2$ is equal to _____

56. Let $a_k = {}^nC_k$ for $0 \le k \le n$ and $A_k = \begin{bmatrix} a_{k-1} & 0 \\ 0 & a_k \end{bmatrix}$ & $B = \sum_{k=1}^{n-1} A_k \cdot A_{k+1} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, then $\frac{a}{b}$ is equal to _____

57. If $\sum_{r=0}^{203} \left(\left(r^2 + 2\right)(r+1)! + 2r(r+1)! \right) = a! - 2(b!)$ (where $a, b \in N$), then $a - b$ is equal to _____

58. If $a = 1 + \frac{x^3}{3!} + \frac{x^6}{6!} + \frac{x^9}{9!} + ..., b = x + \frac{x^4}{4!} + \frac{x^7}{7!} + \frac{x^{10}}{10!} + ...$ and $c = \frac{x^2}{2!} + \frac{x^5}{5!} + \frac{x^8}{8!} + \frac{x^{11}}{11!} + ...$ then the value of $a^3 + b^3 + c^3 - 3abc$ is equal to _____

59. Let $A = \begin{bmatrix} \ell & m & n \\ p & q & r \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^2$. If $(\ell - m)^2 + (p - q)^2 = 9$, $(m - n)^2 + (q - r)^2 = 16$, $(n - \ell)^2 + (r - p)^2 = 25$, then the value of $|\det(B) - 140|$ is _____

60. If the number of ordered pairs (a, b) where $a, b \in R$ such that $(a + ib)^{5015} = (a - ib)^3$ is k, then the unit digit of k is equal to _____

Space for rough work